

On the velocity moments of the lattice Boltzmann equilibrium: second- and third-order results, and the structural cubic defect of the D3Q27 lattice

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Abstract

The discrete velocity moments of the lattice Boltzmann (LB) equilibrium are frequently quoted in two superficially contradictory forms in the literature: a kinetic-theory form that carries the cubic velocity term $\rho u_\alpha u_\beta u_\gamma$ at third order, and a discrete form that does not. We resolve this apparent contradiction by deriving, from first principles and in full, four distinct objects: the second- and third-order moments of (i) the continuous Maxwell–Boltzmann distribution and (ii) the standard second-order Hermite-truncated discrete equilibrium. We show that the two frameworks agree *exactly* at second order, whereas at third order the discrete equilibrium omits precisely $\rho u_\alpha u_\beta u_\gamma$. The algebraic origin of the omission is identified as the parity of the highest lattice moment entering the quadratic part of the equilibrium. We then analyse the D3Q27 lattice component by component and prove that the fully diagonal third-order moment $\Pi_{\alpha\alpha\alpha}$ is locked to ρu_α by the cubic identity $c_{i\alpha}^3 = c_{i\alpha}$, independently of the equilibrium; consequently the diagonal cubic term cannot be restored by *any* Hermite extension on a three-speed lattice, while the mixed components are recoverable. The result is connected to the $O(\text{Ma}^3)$ Galilean-invariance error of standard LB and to its resolution through the cumulant collision operator, whose equilibrium possesses only a finite number of non-vanishing cumulants.

Keywords: lattice Boltzmann method, equilibrium distribution, velocity moments, Hermite expansion, Galilean invariance, cumulant collision operator, D3Q27

1. Introduction

The lattice Boltzmann (LB) method recovers the Navier–Stokes equations through the velocity moments of a discrete equilibrium distribution f_i^{eq} defined on a finite velocity set $\{\mathbf{c}_i\}_{i=0}^{Q-1}$ with weights $\{w_i\}$ [9, 6, 7]. Because the hydrodynamic stress is built from the second- and third-order moments of f_i^{eq} , the precise form of these moments determines which macroscopic equations the scheme realises and at what order in the Mach number.

A recurring source of confusion is that the third-order moment is reported in two different forms. Derivations that begin from continuous kinetic theory (e.g. 6, 10) list the third moment of the *full* Maxwell–Boltzmann distribution, which contains the cubic term $\rho u_\alpha u_\beta u_\gamma$. Treatments that report the moment actually realised by the standard second-order discrete equilibrium (e.g. the textbook account of 7) omit this term. The two statements are not in conflict: they describe different objects. The purpose of this note is to make that distinction completely explicit by deriving all four quantities — the second- and third-order moments of the continuous Maxwellian and of the discrete equilibrium — and then to examine what the D3Q27 lattice can and cannot represent.

The note is organised as follows. Section 2 fixes the continuous and discrete frameworks and states the two sets of moment identities that every subsequent calculation reduces to. Section 3 treats the second-order moment and establishes exact agreement between the two frameworks. Section 4 treats the third-order moment and isolates the cubic discrepancy. Section 6 analyses the D3Q27 lattice component by component. Section 7 collects the consequences for Galilean invariance and for the cumulant collision operator.

2. Preliminaries

2.1. Continuous framework

The continuous Maxwell–Boltzmann equilibrium in D dimensions is

$$f^{\text{eq}}(\boldsymbol{\xi}) = \frac{\rho}{(2\pi c_s^2)^{D/2}} \exp\left(-\frac{|\boldsymbol{\xi} - \mathbf{u}|^2}{2c_s^2}\right), \quad (1)$$

where $\boldsymbol{\xi}$ is the microscopic velocity, ρ the density, $\mathbf{u}$ the macroscopic velocity and $c_s^2 = RT$ the isothermal sound speed squared. Define the ensemble average $\langle \phi \rangle \equiv \int \phi f^{\text{eq}} d^D \boldsymbol{\xi}$ and the peculiar velocity $\mathbf{C} = \boldsymbol{\xi} - \mathbf{u}$, so that under $\mathbf{C}$ the distribution (1) is a zero-mean isotropic Gaussian of variance c_s^2 per component.

Lemma 1 (Gaussian central moments). *With the conventions above,*

$$\langle 1 \rangle = \rho, \quad \langle C_\alpha \rangle = 0, \quad \langle C_\alpha C_\beta \rangle = \rho c_s^2 \delta_{\alpha\beta}, \quad (2)$$

$$\langle C_\alpha C_\beta C_\gamma \rangle = 0, \quad \langle C_\alpha C_\beta C_\gamma C_\delta \rangle = \rho c_s^4 \Delta_{\alpha\beta\gamma\delta}, \quad (3)$$

with $\Delta_{\alpha\beta\gamma\delta} = \delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}$; all odd-order central moments vanish.

Proof. The Gaussian factorises over components, so a multi-index central moment equals $\rho \prod_\mu I_{n_\mu}$, where n_μ is the number of repetitions of direction μ and $I_n = \int x^n (2\pi c_s^2)^{-1/2} e^{-x^2/2c_s^2} dx$ with $I_0 = 1$, $I_1 = 0$, $I_2 = c_s^2$, $I_3 = 0$, $I_4 = 3c_s^4$. Any factor with odd n_μ annihilates the product, giving the stated odd-order vanishing and (2). For the fourth-order moment the surviving index patterns are the three pairings, each contributing $I_2 I_2 = c_s^4$, which is the Isserlis/Wick result $\rho c_s^4 \Delta_{\alpha\beta\gamma\delta}$. $\square$

2.2. Discrete framework

The standard LB equilibrium is the second-order Hermite truncation of (1) evaluated on the lattice [9, 6, 10],

$$f_i^{\text{eq}} = w_i \rho \left[1 + \frac{c_{i\mu} u_\mu}{c_s^2} + \frac{u_\mu u_\nu (c_{i\mu} c_{i\nu} - c_s^2 \delta_{\mu\nu})}{2c_s^4} \right]. \quad (4)$$

Standard isothermal velocity sets (D2Q9, D3Q15, D3Q19, D3Q27) are Gauss-Hermite quadratures exact to polynomial degree five [6, 10], so the weights satisfy the moment identities

$$\sum_i w_i = 1, \quad \sum_i w_i c_{i\alpha} = 0, \quad (5a)$$

$$\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}, \quad \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} = 0, \quad (5b)$$

$$\sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} = c_s^4 \Delta_{\alpha\beta\gamma\delta}, \quad \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} c_{i\epsilon} = 0. \quad (5c)$$

Equations (5a)–(5c) are labelled $M_0, \dots, M_5$ below. The macroscopic moments are $\sum_i f_i^{\text{eq}} = \rho$ and $\sum_i f_i^{\text{eq}} c_{i\alpha} = \rho u_\alpha$ by construction.

The third moment of the third-order equilibrium treated in Section 5 requires one further identity, the sixth-order weight moment, which is the *first* member of the hierarchy that departs from full isotropy. For the tensor-product lattices it reads

$$M_{\alpha\beta\gamma\delta\epsilon\zeta}^{(6)} \equiv \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} c_{i\epsilon} c_{i\zeta} = c_s^6 \Delta_{\alpha\beta\gamma\delta\epsilon\zeta}^{(6)} + A \delta_{\alpha\beta\gamma\delta\epsilon\zeta}^{[6]}, \quad (6)$$

where $\Delta^{(6)}$ is the fully symmetric sum of the $5!! = 15$ pairings of six indices into three Kronecker deltas, the diagonal indicator $\delta_{\alpha\beta\gamma\delta\epsilon\zeta}^{[6]}$ equals unity iff all six indices coincide and zero otherwise, and the anisotropic defect amplitude is $A = m_6 - 15c_s^6$ with $m_6 = \sum_i w_i c_{ix}^6$ the one-dimensional sixth moment. On D3Q27 the cubic identity $c_{ix}^6 = c_{ix}^2$ gives $m_6 = c_s^2 = \frac{1}{3}$, against the isotropic value $15c_s^6 = \frac{5}{9}$, so $A = -\frac{2}{9}$. The fourth-order moment, by contrast, is fully isotropic ($m_4 = 3c_s^4$, consistent with (5c)); thus M_6 is the lowest even moment to break isotropy, and the breakage is confined to the fully diagonal entries $M_{xxxxxx}^{(6)} = m_6$.

3. The second-order moment

Proposition 1 (Continuous second moment). $\int f^{\text{eq}} \xi_\alpha \xi_\beta d^D \xi = \rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}$.

Proof. Writing $\xi_\alpha = u_\alpha + C_\alpha$ and using Lemma 1,

$$\langle \xi_\alpha \xi_\beta \rangle = u_\alpha u_\beta \langle 1 \rangle + u_\alpha \langle C_\beta \rangle + u_\beta \langle C_\alpha \rangle + \langle C_\alpha C_\beta \rangle = \rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}. \quad \square$$

Proposition 2 (Discrete second moment). $\sum_i f_i^{\text{eq}} c_{i\alpha} c_{i\beta} = \rho c_s^2 \delta_{\alpha\beta} + \rho u_\alpha u_\beta$.

Proof. Insert (4) and split into the three terms of the bracket. *Constant term:* by M_2 , $\rho \sum_i w_i c_{i\alpha} c_{i\beta} = \rho c_s^2 \delta_{\alpha\beta}$. *Linear term:* by M_3 , $(\rho u_\mu / c_s^2) \sum_i w_i c_{i\mu} c_{i\alpha} c_{i\beta} = 0$. *Quadratic term:* by M_4 and M_2 ,

$$\begin{aligned} \frac{\rho u_\mu u_\nu}{2c_s^4} \left[\sum_i w_i c_{i\mu} c_{i\nu} c_{i\alpha} c_{i\beta} - c_s^2 \delta_{\mu\nu} \sum_i w_i c_{i\alpha} c_{i\beta} \right] &= \frac{\rho u_\mu u_\nu}{2} [\delta_{\mu\nu} \delta_{\alpha\beta} + \delta_{\mu\alpha} \delta_{\nu\beta} + \delta_{\mu\beta} \delta_{\nu\alpha} - \delta_{\mu\nu} \delta_{\alpha\beta}] \\ &= \frac{\rho}{2} (u_\alpha u_\beta + u_\beta u_\alpha) = \rho u_\alpha u_\beta. \end{aligned}$$

Summation gives the stated result. $\square$

Remark 1. *Propositions 1 and 2 coincide: the second-order moment, i.e. the momentum-flux tensor $\Pi_{\alpha\beta} = \rho u_\alpha u_\beta + p \delta_{\alpha\beta}$ with $p = \rho c_s^2$, is reproduced exactly by the second-order discrete equilibrium. This exactness is what allows standard LB to recover the viscous stress correctly at leading order [7].*

4. The third-order moment

Proposition 3 (Continuous third moment).

$$\int f^{\text{eq}} \xi_\alpha \xi_\beta \xi_\gamma d^D \xi = \rho c_s^2 (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}) + \rho u_\alpha u_\beta u_\gamma. \quad (7)$$

Proof. Expand $\xi_\alpha \xi_\beta \xi_\gamma = (u_\alpha + C_\alpha)(u_\beta + C_\beta)(u_\gamma + C_\gamma)$ and average term by term with Lemma 1. The triple-mean term yields $\rho u_\alpha u_\beta u_\gamma$ (it survives because $\langle 1 \rangle = \rho$); the three terms linear in $\mathbf{C}$ vanish through $\langle C \rangle = 0$; the three terms with two peculiar factors give $u_\alpha \langle C_\beta C_\gamma \rangle + (\text{perm.}) = \rho c_s^2 (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta})$; and the triple-peculiar term vanishes as an odd Gaussian moment. $\square$

Proposition 4 (Discrete third moment).

$$\sum_i f_i^{\text{eq}} c_{i\alpha} c_{i\beta} c_{i\gamma} = \rho c_s^2 (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}). \quad (8)$$

Proof. Insert (4). *Constant term:* by M_3 , $\rho \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} = 0$. *Linear term:* by M_4 ,

$$\frac{\rho u_\mu}{c_s^2} \sum_i w_i c_{i\mu} c_{i\alpha} c_{i\beta} c_{i\gamma} = \frac{\rho u_\mu}{c_s^2} c_s^4 (\delta_{\mu\alpha} \delta_{\beta\gamma} + \delta_{\mu\beta} \delta_{\alpha\gamma} + \delta_{\mu\gamma} \delta_{\alpha\beta}) = \rho c_s^2 (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}).$$

Quadratic term: the leading contraction is the fifth-order lattice moment, which vanishes by M_5 , and the subtracted piece vanishes by M_3 ; hence the quadratic term contributes zero. Summation gives (8). $\square$

Remark 2 (The cubic discrepancy). *Subtracting (8) from (7),*

$$\underbrace{\int f^{\text{eq}} \xi_\alpha \xi_\beta \xi_\gamma d^D \xi}_{\text{continuous}} - \underbrace{\sum_i f_i^{\text{eq}} c_{i\alpha} c_{i\beta} c_{i\gamma}}_{\text{discrete } O(u^2)} = \rho u_\alpha u_\beta u_\gamma. \quad (9)$$

The structural reason for the omission is a parity argument visible in the two proofs. In the second moment (Proposition 2) the highest lattice moment reached by the quadratic part of the equilibrium is the even M_4 , which is non-zero and supplies $\rho u_\alpha u_\beta$. In the third moment the corresponding contraction reaches the odd M_5 , which vanishes; the cubic term therefore has no source in a second-order equilibrium. This missing contribution is the well-known nonlinear ($O(\text{Ma}^3)$) deviation of the lattice BGK models from the Navier–Stokes stress [8, 7].

5. The third-order equilibrium and its discrete third moment

Remark 2 located the cubic discrepancy in the truncation of the equilibrium at second order. The natural remedy is to retain the next term of the Hermite expansion of (1). We write this third-order equilibrium explicitly,

then recompute the discrete third moment using the sixth-order identity (6). The cubic term is recovered for every mixed index triple but not for the fully diagonal ones; the latter failure is shown, in Section 6, to be a property of the velocity set independent of the equilibrium.

5.1. The third-order Hermite equilibrium

The expansion of the Maxwellian (1) retained to third Hermite order is [5, 6, 10]

$$f_i^{\text{eq},(3)} = w_i \rho \left[1 + \frac{c_{i\mu} u_\mu}{c_s^2} + \frac{u_\mu u_\nu \mathcal{H}_{i,\mu\nu}^{(2)}}{2c_s^4} + \frac{u_\mu u_\nu u_\lambda \mathcal{H}_{i,\mu\nu\lambda}^{(3)}}{6c_s^6} \right], \quad (10)$$

with the second- and third-order tensor Hermite polynomials

$$\mathcal{H}_{i,\mu\nu}^{(2)} = c_{i\mu} c_{i\nu} - c_s^2 \delta_{\mu\nu}, \quad (11)$$

$$\mathcal{H}_{i,\mu\nu\lambda}^{(3)} = c_{i\mu} c_{i\nu} c_{i\lambda} - c_s^2 (c_{i\mu} \delta_{\nu\lambda} + c_{i\nu} \delta_{\mu\lambda} + c_{i\lambda} \delta_{\mu\nu}). \quad (12)$$

By index class, the ten independent components of (12) in three dimensions are

$$\underbrace{\mathcal{H}_{xxx}^{(3)} = c_x^3 - 3c_s^2 c_x}_{\text{diagonal}}, \quad \underbrace{\mathcal{H}_{xxy}^{(3)} = (c_x^2 - c_s^2) c_y}_{\text{two equal}}, \quad \underbrace{\mathcal{H}_{xyz}^{(3)} = c_x c_y c_z}_{\text{all distinct}}, \quad (13)$$

together with their index permutations (three diagonal, six two-equal, one all-distinct).

Remark 3 (Vanishing of the diagonal Hermite mode on D3Q27). *On D3Q27 the cubic identity $c_{ix}^3 = c_{ix}$ together with $3c_s^2 = 1$ give*

$$\mathcal{H}_{i,xxx}^{(3)}|_{D3Q27} = c_{ix}^3 - 3c_s^2 c_{ix} = c_{ix} - c_{ix} = 0 \quad (14)$$

identically over the lattice. The third-order Hermite subspace represented by D3Q27 thus contains only the seven mixed modes; the three diagonal modes are the zero vector. No equilibrium can populate a basis direction absent from the lattice.

5.2. Discrete third moment of the third-order equilibrium

Proposition 5 (Discrete third moment under $f_i^{\text{eq},(3)}$). *On D3Q27, the third moment of the third-order equilibrium (10) is*

$$\sum_i f_i^{\text{eq},(3)} c_{i\alpha} c_{i\beta} c_{i\gamma} = \rho c_s^2 (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}) + \rho u_\alpha u_\beta u_\gamma (1 - \delta_{\alpha\beta\gamma}^{[3]}), \quad (15)$$

where $\delta_{\alpha\beta\gamma}^{[3]} = 1$ if $\alpha = \beta = \gamma$ and zero otherwise. The cubic term is recovered exactly for every mixed index triple and is absent for the fully diagonal components.

Proof. Write $\Pi_{\alpha\beta\gamma} = \Pi_{\alpha\beta\gamma}^{(2)} + \Delta\Pi_{\alpha\beta\gamma}$, where $\Pi^{(2)}$ collects the first three terms of (10) and equals $\rho c_s^2(u_\alpha\delta_{\beta\gamma} + u_\beta\delta_{\alpha\gamma} + u_\gamma\delta_{\alpha\beta})$ by Proposition 4, and the third-order term contributes

$$\Delta\Pi_{\alpha\beta\gamma} = \frac{\rho}{6c_s^6} u_\mu u_\nu u_\lambda \Theta_{\mu\nu\lambda, \alpha\beta\gamma}, \quad \Theta_{\mu\nu\lambda, \alpha\beta\gamma} = \sum_i w_i \mathcal{H}_{i, \mu\nu\lambda}^{(3)} c_{i\alpha} c_{i\beta} c_{i\gamma}.$$

The inverse Hermite relation $c_{i\alpha} c_{i\beta} c_{i\gamma} = \mathcal{H}_{i, \alpha\beta\gamma}^{(3)} + c_s^2(\delta_{\alpha\beta} c_{i\gamma} + \delta_{\alpha\gamma} c_{i\beta} + \delta_{\beta\gamma} c_{i\alpha})$ splits Θ into a product of two third-order Hermite polynomials plus cross terms $\sum_i w_i \mathcal{H}_{i, \mu\nu\lambda}^{(3)} c_{i\gamma}$. The cross terms are degree-four polynomials, integrated exactly by the fifth-degree quadrature, and therefore equal their continuum value $\langle \mathcal{H}^{(3)} \mathcal{H}^{(1)} \rangle = 0$ by Hermite orthogonality. Hence

$$\Theta_{\mu\nu\lambda, \alpha\beta\gamma} = \sum_i w_i \mathcal{H}_{i, \mu\nu\lambda}^{(3)} \mathcal{H}_{i, \alpha\beta\gamma}^{(3)}.$$

This degree-six sum equals its continuum value $c_s^6 P_{\mu\nu\lambda, \alpha\beta\gamma}$, with $P = \sum_{\sigma \in S_3} \delta_{\mu\sigma_1} \delta_{\nu\sigma_2} \delta_{\lambda\sigma_3}$ the symmetric pairing of $(\mu\nu\lambda)$ with $(\alpha\beta\gamma)$, provided no monomial c_{ix}^6 arises — the only entry of (6) that carries the defect A . Such a monomial requires both factors to supply c_{ix}^3 , that is $\mu = \nu = \lambda = x$ and $\alpha = \beta = \gamma = x$.

Mixed indices (α, β, γ not all equal): $\mathcal{H}_{\alpha\beta\gamma}^{(3)}$ carries no c_{ix}^3 term, so no c_{ix}^6 arises for any $(\mu\nu\lambda)$ and the sum is exact. Contracting with $u_\mu u_\nu u_\lambda$, each permutation in P contributes $u_\alpha u_\beta u_\gamma$, so $u_\mu u_\nu u_\lambda \Theta = 6c_s^6 u_\alpha u_\beta u_\gamma$ and $\Delta\Pi_{\alpha\beta\gamma} = \rho u_\alpha u_\beta u_\gamma$.

Diagonal indices ($\alpha = \beta = \gamma = x$): by Remark 3, $\mathcal{H}_{i, xxx}^{(3)} \equiv 0$ over D3Q27, so the correction associated with this mode in (10) is identically zero and contributes nothing, giving $\Delta\Pi_{xxx} = 0$. Section 6 reaches the same conclusion through the quadrature defect A of (6) and through an equilibrium-independent identity. Collecting both cases yields (15). $\square$

A concrete check of the mixed case is instructive. For Π_{xyz} with x, y, z distinct, the second-order part vanishes; on the tensor-product lattice the only non-vanishing internal indices are the permutations of (x, y, z) , each yielding $\sum_i w_i c_{ix}^2 c_{iy}^2 c_{iz}^2 = m_2^3 = c_s^6$. Summing the six permutations, $\Delta\Pi_{xyz} = (\rho/6c_s^6) \cdot 6c_s^6 u_x u_y u_z = \rho u_x u_y u_z$, in agreement with the continuum value (7).

6. Component-wise reality on the D3Q27 lattice

Proposition 5 established the recovery pattern by direct computation with the third-order equilibrium. We now give an equilibrium-independent argument that reaches the same conclusion for the diagonal components, exposing it as a property of the velocity set alone, and collect the result in a single table. Throughout this section we use the D3Q27 lattice in lattice units, $c_{i\alpha} \in \{-1, 0, 1\}$, $c_s^2 = 1/3$.

6.1. The diagonal lock

Proposition 6 (Diagonal cubic lock). *On any lattice with $c_{i\alpha} \in \{-1, 0, 1\}$ (D2Q9, D3Q15, D3Q19, D3Q27),*

$$\Pi_{\alpha\alpha\alpha} \equiv \sum_i f_i c_{i\alpha}^3 = \rho u_\alpha \quad \text{for every } f_i, \quad (16)$$

in particular independently of the choice or Hermite order of the equilibrium.

Proof. Since $c_{i\alpha} \in \{-1, 0, 1\}$, the cubic identity $c_{i\alpha}^3 = c_{i\alpha}$ holds pointwise. Hence $\sum_i f_i c_{i\alpha}^3 = \sum_i f_i c_{i\alpha} = \rho u_\alpha$ for any distribution. $\square$

Because the continuous target (7) requires $\Pi_{\alpha\alpha\alpha} = 3\rho c_s^2 u_\alpha + \rho u_\alpha^3 = \rho u_\alpha + \rho u_\alpha^3$ (using $3c_s^2 = 1$), the diagonal cubic term ρu_α^3 has no degree of freedom on a three-speed lattice and cannot be represented.

6.2. The sixth-order moment defect

The same obstruction appears, equivalently, as a quadrature defect. Augmenting (4) with the third-order Hermite term and evaluating its contribution to the diagonal moment Π_{xxx} reduces, on the one-dimensional factor D1Q3, to

$$\frac{u_x^3}{6c_s^6}(M_6 - 3c_s^2 M_4), \quad M_6 = \sum_i w_i c_{ix}^6, \quad M_4 = \sum_i w_i c_{ix}^4. \quad (17)$$

For D1Q3 one has $M_4 = \frac{1}{3}$ (which equals $3c_s^4$, so the fourth-order identity in (5c) holds), whereas $M_6 = \frac{1}{3} \neq 15c_s^6 = \frac{5}{9}$. Substituting, $M_6 - 3c_s^2 M_4 = \frac{1}{3} - 3 \cdot \frac{1}{3} \cdot \frac{1}{3} = 0$, so the third-order Hermite term contributes *exactly zero* to the diagonal moment: the lattice sixth-order defect cancels the very term it was introduced to supply. This is the quadrature-level statement of Proposition 6, and is consistent with the lattice's fifth-degree quadrature accuracy [10].

Table 1: Third-order equilibrium moment on D3Q27 ($c_s^2 = 1/3$, $3c_s^2 = 1$), per unit ρ , for the three index classes. The second-order column is Proposition 4; the Hermite-3 column is the best attainable on a three-speed lattice; the Maxwell column is the continuum target (7).

Component	Continuous Maxwell	Discrete $O(u^2)$	Discrete + Hermite-3
Π_{xxx} (diagonal)	$u_x + u_x^3$	u_x	u_x (locked)
Π_{xxy} (two equal)	$\frac{1}{3}u_y + u_x^2u_y$	$\frac{1}{3}u_y$	$\frac{1}{3}u_y + u_x^2u_y$
Π_{xyz} (all distinct)	$u_xu_yu_z$	0	$u_xu_yu_z$

6.3. Recoverable mixed components

For a component in which no single direction carries a power exceeding four, the relevant lattice moment is a tensor product of one-dimensional factors, each exact, so the third-order Hermite term *does* restore the cubic contribution. For example $\sum_i w_i c_{ix}^2 c_{iy}^2 c_{iz}^2 = c_s^6$ matches the continuum, and the mixed moments are recovered:

$$\Pi_{xyz}^{\text{cont}} = \rho u_x u_y u_z, \quad \Pi_{xxy}^{\text{cont}} = \rho c_s^2 u_y + \rho u_x^2 u_y, \quad (18)$$

both attainable with the third-order Hermite equilibrium. Only the fully diagonal components remain locked. Table 1 collects the three representative cases.

7. Consequences and concluding remarks

The three rows of Table 1 establish a strict ordering of representable third moments,

$$\underbrace{\rho c_s^2(\cdots)}_{O(u^2) \text{ truncation}} \subsetneq \underbrace{\rho c_s^2(\cdots) + \rho u_\alpha u_\beta u_\gamma}_{\text{Hermite-3 on D3Q27}} \Big|_{\text{mixed only}} \subsetneq \underbrace{\rho c_s^2(\cdots) + \rho u_\alpha u_\beta u_\gamma}_{\text{continuous Maxwell}} \Big|_{\text{all}}. \quad (19)$$

The gap between the second and third members is the diagonal cubic term, which the three-speed lattice cannot carry. This is precisely the lattice-induced $O(\text{Ma}^3)$ violation of Galilean invariance of the standard scheme [8, 2]; its complete removal requires multi-speed (higher-quadrature) velocity sets such as those constructed from fifth-degree Gauss–Hermite rules [3].

It is instructive to see why the cumulant collision operator [4] sidesteps the issue rather than attempting to restore the diagonal raw moment, which Proposition 6 shows to be impossible. The two-sided Laplace transform of (1) is $M(\mathbf{k})/\rho = \exp(k_\mu u_\mu + \frac{1}{2}c_s^2 k_\mu k_\mu)$, so the cumulant generating function

$\ln(M/\rho)$ is a second-degree polynomial: the Maxwellian possesses only a first cumulant $\kappa_{1,\alpha} = u_\alpha$ and a second cumulant $\kappa_{2,\alpha\beta} = c_s^2 \delta_{\alpha\beta}$, with all higher cumulants identically zero [4]. Through the moment–cumulant relation the cubic term in (7) is the reducible contribution $\kappa_{1,\alpha}\kappa_{1,\beta}\kappa_{1,\gamma} = \rho u_\alpha u_\beta u_\gamma$, which is frame-dependent rather than a genuine third-order cumulant. The cumulant operator relaxes the Galilean-invariant cumulants directly, setting the third-order cumulant to its (vanishing) equilibrium value; it therefore neither requires nor falsely advertises recovery of the diagonal raw moment, which accounts for its improved Galilean behaviour at high Reynolds number relative to a Hermite-extended BGK equilibrium on the same D3Q27 lattice.

In summary, the apparently conflicting third-order results in the literature are two correct statements about two different objects: the continuum Maxwellian carries $\rho u_\alpha u_\beta u_\gamma$ (Proposition 3), the second-order discrete equilibrium does not (Proposition 4), and on the D3Q27 lattice the diagonal portion of that term is structurally unrepresentable (Proposition 6). The second-order moment, by contrast, is exact in both frameworks (Propositions 1–2), which is why standard LB recovers the Navier–Stokes stress at the intended order.

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